

Consensus Consolidation Turns Correct Shared Experience into Silent Strategy Degradation in Multi-Agent LLM Systems

Phase-Beta results section (draft)

July 10, 2026

Draft status. All numbers below come from the preregistered Phase-Beta protocol (`prereg_phase_beta.md`); gate rules were frozen before data collection and each verdict is recorded in that document’s deviation log. Statistics are same-maze paired bootstrap (10,000 resamples, 95% CI). Experience text is entirely model-generated; the sanitizer guarantees no researcher-authored string enters the memory pool.

Abstract

We study whether shared experiential memory in multi-agent LLM systems can turn locally correct strategies into globally degraded behavior. Using a controlled maze microscope where every action is legal and shortest-path cost is measurable, we run a preregistered ladder of experiments with faithful instantiations of ExpeL-style insight operations and Generative-Agents-style retrieval. We find: (i) at the single-agent level, faithful experience learning does not measurably change the primary efficiency endpoints (an attribution control); (ii) at the multi-agent level, injecting experience degrades efficiency and amplifies loops, with *consensus-consolidated* shared memory producing the most reproducible degradation across seeds and the only significant loop amplification relative to private memory; (iii) the harm does not come from read-side usage-weighted retrieval (a null dose-response), but from write-side consolidation, which compresses the population’s active strategy supply by roughly a factor of two. The failure is silent: agents still succeed, never cross walls, and never falsely submit, yet increasingly walk in circles.

1 Experimental Ladder and Verdicts

We evaluate on a generated 15×15 trap-maze family (shortest path ≥ 30) with local observations, a state-guided controller, and a step cap of 120. Solvers never see the maze graph or shortest path. Memory conditions differ only in *write protocol* and *sharing*; the reviewer sees agent logs and quality signals, never hidden routes. The primary endpoints, frozen in advance, are success-only excess steps, loop rate, and the multi-agent ant-mill rate.

1.1 E1: single-agent attribution control

Before introducing sharing, we ask whether faithful experience learning changes single-agent behavior at all. Relative to a no-memory baseline (paired, final round, three seeds), neither the ExpeL-operations writer nor the append writer moves any primary endpoint: success-only excess steps CIs contain zero (append -4.74 , CI $[-14.1, +4.2]$; ops $+5.11$, CI $[-6.4, +16.3]$), as do loop rates. By the frozen rule this is a **FAIL**. However, the mandated adherence analysis shows experience is *not* ignored: the ops writer shifts action selection away from the controller suggestion after memory activates ($+0.041$ deviation, same sign in 3/3 seeds), and the append writer significantly

reduces maximum revisits. The single-agent null is therefore an *attribution control*: any multi-agent degradation cannot be attributed to “using memory” per se, only to the sharing channel.

1.2 E2: multi-agent memory matrix

We compare five arms at four agents, $T = 6$, five seeds, against a write-only *frozen* control that accumulates but never injects experience. Table 1 and Figure 2 report paired differences at the final round.

Table 1: E2 final-round paired differences versus frozen write-only memory (five seeds, $n_{\text{pairs}} = 240$; success-only excess steps drops unpaired failures). * = 95% CI excludes zero.

Arm	Success	Success excess steps	Loop rate	Max revisits
MAS no memory	−0.004	+1.12	+0.042*	+0.10
Private memory	−0.175*	+7.32*	+0.079*	+1.20*
Shared append	−0.121*	+2.75	+0.025	+0.23
Shared consolidated	−0.171*	+11.84*	+0.138*	+1.68*

Three readings follow. First, multi-agent execution is not itself pathological: MAS-no-memory matches frozen on efficiency and success, with only a small loop increase (+0.042, CI [0.008, 0.075]). Second, **consensus-consolidated shared memory meets every preregistered degradation criterion**: success-only excess steps +11.84 (CI [7.66, 16.02]), same-sign in 5/5 seeds and *larger* in the later seeds; loop rate +0.138 (CI [0.092, 0.188]); success −0.171. Third, the broad risk is active injection rather than sharing alone—private memory also degrades—so we test the sharing-specific claim directly.

Shared versus private boundary. Figure 3 contrasts the shared arms against a *private* baseline. The efficiency separation is not significant (consolidated vs private success-only excess steps +0.87, CI [−4.29, +6.15]), but loop amplification is: consolidated loops significantly more than private (+0.058, CI [0.008, 0.113]), while append loops *less*. The honest claim is therefore that consensus consolidation specifically amplifies looping relative to private memory; we do not claim a broad shared-over-private efficiency gap.

1.3 E3: read-side dose-response is null

If the harm were read-side—agents increasingly retrieving a few high-vote memories—then raising the importance weight λ in Generative-Agents-style scoring should deepen degradation. It does not. Sweeping $\lambda \in \{0, 0.5, 1\}$ on the append arm (three seeds, $T = 6$), the paired difference of $\lambda=1$ versus $\lambda=0$ on success-only excess steps is +1.69 (CI [−4.26, +7.61]), loop rate +0.049 (CI [0.000, 0.104]), and final-round retrieval entropy is non-monotone in λ (0.833/0.826/0.845). The read-side usage-weighted-feedback hypothesis is **refuted**; we report it as a clean null.

1.4 Mechanism: write-side strategy-supply compression

An offline audit of the same E2 runs locates the concentration on the *write* side (Figure 4). Under identical settings, the consolidated pool converges to 14.2 items with only 11.0 distinct experiences ever injected across all rounds, whereas the append pool grows to the cap (80) and injects 25.4 distinct experiences, and private injects 37.6. Consensus consolidation compresses the population’s

active strategy supply by roughly $2.3\times$ relative to append. The top-1 retrieval share rises in step, but this is a consequence of the smaller pool rather than an independent channel; the load-bearing evidence is the direct pool-size and distinct-injection counts.

1.5 E4: long-horizon stress test (exploratory)

To ask whether the write-side harm saturates, deepens, or self-heals over a longer feedback horizon, we extend the four-agent shared arms to $T=10$ (three seeds, heldout 12, frozen and shared-append included as controls; not gate-bound, per the preregistration). Relative to frozen, shared-consolidated’s degradation is present at every checkpoint we tested ($t = 3, 6, 9$) and does not narrow: success drops -0.174 , -0.146 , -0.257 (all CIs exclude zero), and loop rate rises $+0.083$, $+0.083$, $+0.118$ (all CIs exclude zero), with the final round the worst of the three. A stricter within-condition test—comparing each condition’s own last three rounds to its first three (excluding the warm-up round $t=0$)—finds no significant self-trend for shared-consolidated on any of six endpoints: the gap opens early and persists at a noisy plateau rather than provably deepening. The append arm, in contrast, shows a significant within-condition *improvement* over the same horizon (cost ratio -0.079 , excess steps -3.50 , stagnation -0.026 , max revisits -0.340 , all CIs exclude zero) and by $t=9$ is significantly cheaper than frozen on cost ratio, excess steps, and max revisits—consistent with a write protocol whose strategy supply is not compressed continuing to benefit from accumulation. Splitting shared-consolidated’s within-condition trend by seed exposes heterogeneity the pooled estimate hides. Seed 1 is the only seed with a significant late-vs-early decline in success (-0.097 , CI excludes zero), but its round-wise trajectory oscillates rather than falling monotonically. Seed 2 has the most severe round-wise trajectory, ending at success 0.17 and loop rate 0.46, although its late-vs-early success CI still includes zero. Seed 0 shows no significant decline. The pooled “no significant further drift” verdict is therefore a mean over a mixed population, not evidence that every instance is equally stable; we flag this as a target for external-validity follow-up rather than a settled claim.

2 The Failure Is Silent

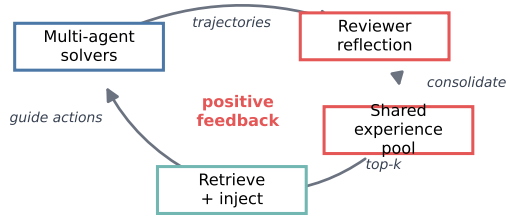
Figure 1 shows the mechanism schematic and a case study mined from the runs: the same held-out maze and agent at the final round, whose shortest path is 38 steps. As the write protocol consolidates, the route degrades monotonically—frozen 39 steps (one revisit), private 47 (four), shared-append 55 (nine), shared-consolidated 73 with cost 1.92, eleven revisits to the same cell, and a detected loop—yet every route is legal and every arm still reaches the goal. This is the ant-mill signature: not a false belief, but a locally plausible correction signal, reinforced through consensus writing, that composes into a globally wasteful cycle.

3 Summary of Claims

- Single-agent faithful experience learning does not move the primary endpoints (attribution control; E1).
- Multi-agent experience injection degrades efficiency and amplifies loops; consensus consolidation is the most reproducible degrading arm across seeds (E2).
- Relative to private memory, the shared-specific effect is loop amplification, not a broad efficiency gap (E2 boundary).
- The harm is not read-side usage-weighted retrieval (null dose-response; E3) but write-side consolidation that compresses the active strategy supply (mechanism audit).
- Extending to a longer horizon ($T=10$), the consolidation harm persists without provable further deepening at the population level. Per-seed decomposition separates one significant late-vs-early decline from a different, more extreme trajectory whose CI remains inconclusive (E4, exploratory).

Figure 1 | Consensus consolidation turns correct shared experience into a silent ant-mill loop

a Shared-experience feedback loop



b Silent degradation on one held-out maze

Same maze, same agent, final round (seed 4). Every route is legal — no wall crossing, no false submit — and every arm still reaches the goal. As the write protocol consolidates experience, the agent revisits more cells and finally loops, while shortest path stays 38 steps.

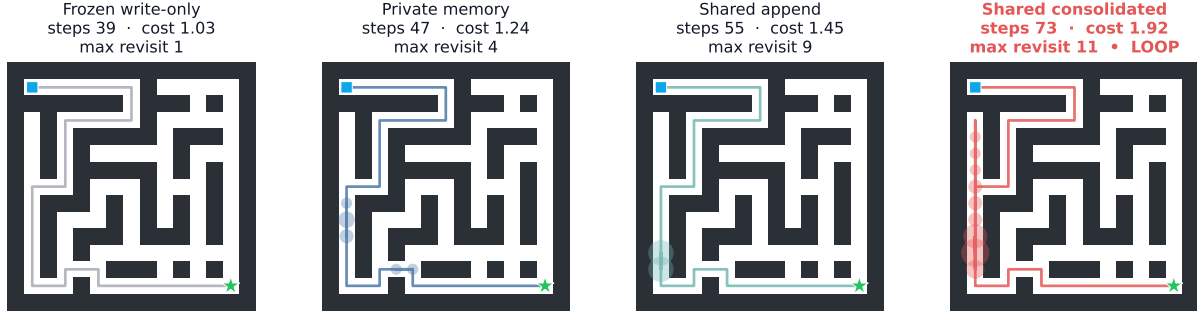


Figure 1: Mechanism microscope and a silent degradation case. (a) The shared-experience feedback loop: solver trajectories are reflected into a shared pool, consolidated by LLM-issued operations, retrieved top- k , and injected back as guidance—a positive-feedback cycle. (b) Same held-out maze and agent, final round; shortest path 38. Revisit heat (shaded blobs) grows as the write protocol consolidates, culminating in a silent loop under shared-consolidated memory, while every route stays legal and successful.

E2: active consolidated memory degrades efficiency and increases loops

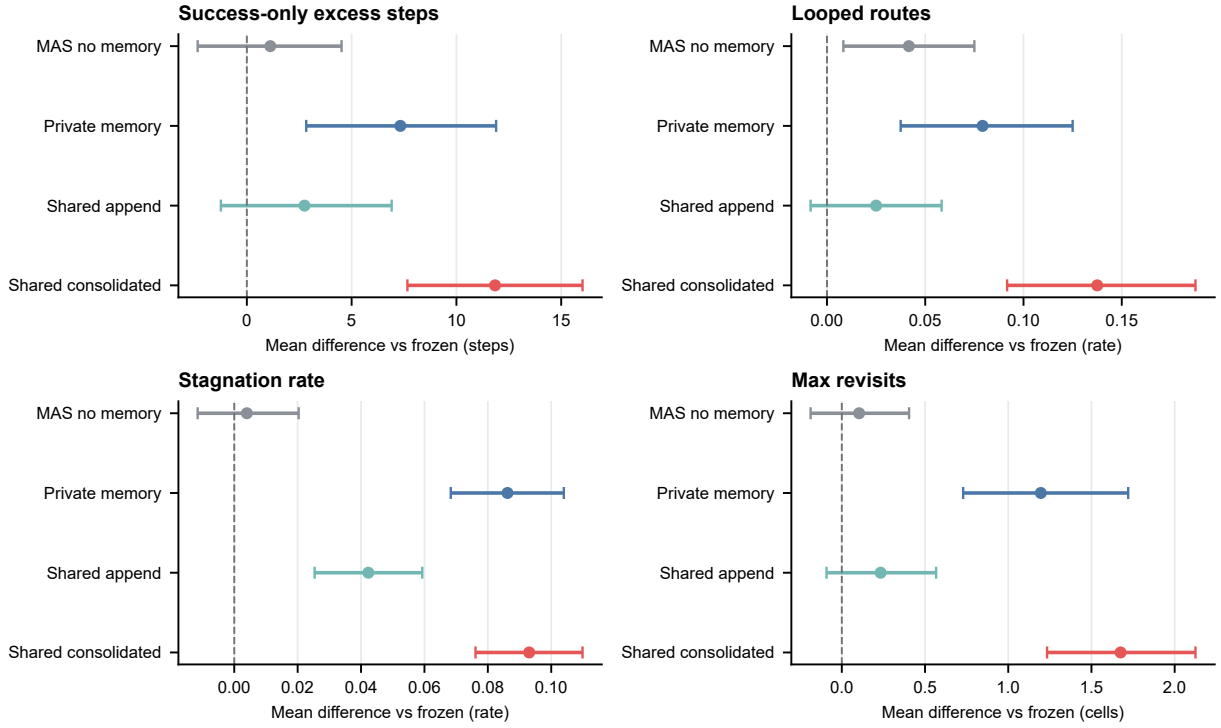


Figure 2: **E2 main result: paired differences versus frozen write-only memory** (five seeds, final round). Points are mean differences; bars are 95% bootstrap CIs; the dashed line is zero. Shared-consolidated (red) degrades on all four endpoints with the largest effects; MAS-no-memory (grey) is on the zero line except for a small loop increase.

E2 boundary: vs private memory, only looping separates (filled = 95% CI excludes 0)

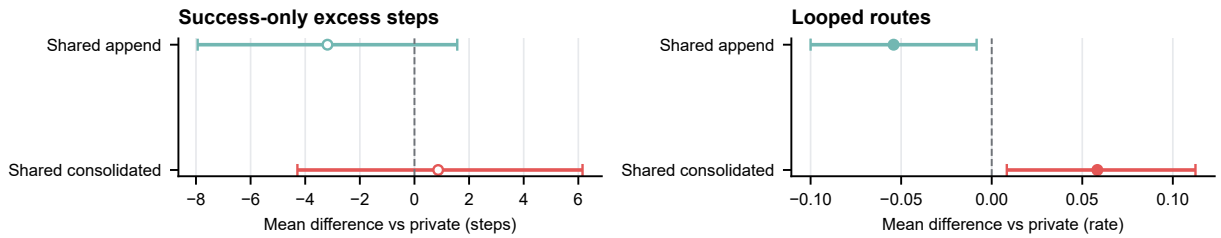


Figure 3: **E2 boundary: shared arms versus a private baseline**. Filled markers mark CIs that exclude zero. Only looping separates shared-consolidated from private (+0.058, CI [0.008, 0.113]); the efficiency contrast does not, and shared-append actually loops less than private.

E2 mechanism audit: consolidation compresses the strategy supply

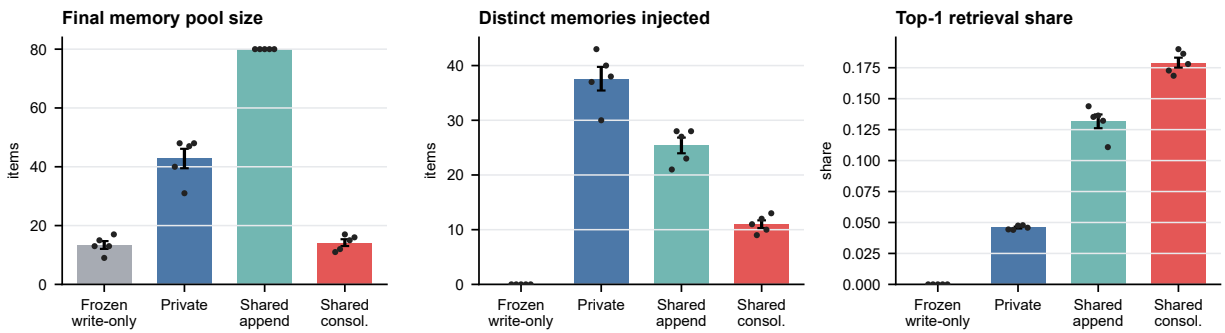


Figure 4: **Mechanism audit: consolidation compresses the strategy supply.** Final pool size, distinct experiences ever injected, and top-1 retrieval share across five seeds (dots). Consensus consolidation keeps a small pool and injects the fewest distinct strategies, concentrating retrieval; the direct counts (left, centre) carry the causal claim.